Chapter 6. Sequences and Series of Functions

Uniform convergence, differentiation and integration of limits, double series, power series, trigonometric series

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6.1 Uniform convergence

Let $X \subset \mathbb{R}^n$ and $f_n, f : X \to \mathbb{R}^m$ (or $\mathbb{C}$). $f_n \to f$ **pointwise** if $f_n(x) \to f(x)$ for each x, and **uniformly** if $\|f_n - f\|_{\sup} = \sup_{x\in X}\|f_n(x) - f(x)\| \to 0$.

Example 6.1

$x^n \to 0$ pointwise on $[0,1)$ but not uniformly, since $\sup_{[0,1)} x^n = 1$; the limit of x^n on $[0,1]$ is discontinuous. $f_n(x) = nx(1-x^2)^n$ tends to 0 pointwise on $[0,1]$, while $\int_0^1 f_n = \frac{n}{2n+2} \to \frac{1}{2}$.

Theorem 6.1 (6.1.1)

If continuous $f_n : X \to Y$ converge uniformly to f, then f is continuous.

Proof. Given x_0 and ϵ, choose N with $\|f_N - f\|_{\sup} < \epsilon/3$ and δ with $\|f_N(x) - f_N(x_0)\| < \epsilon/3$ for $\|x - x_0\| < \delta$. Then

$$\|f(x) - f(x_0)\| \le \|f(x) - f_N(x)\| + \|f_N(x) - f_N(x_0)\| + \|f_N(x_0) - f(x_0)\| < \epsilon.$$

■

Theorem 6.2 (6.1.2, uniform Cauchy criterion)

(f_n) converges uniformly on X iff for every $\epsilon > 0$ there is N with $\|f_m(x) - f_n(x)\| < \epsilon$ for all $m, n \ge N$ and $x \in X$.

Proof. Necessity is the triangle inequality. Conversely, each $(f_n(x))$ is Cauchy, so converges to some $f(x)$ (Theorem 2.14 of Volume I); letting $m \to \infty$ in $\|f_m(x) - f_n(x)\| < \epsilon$ gives $\|f(x) - f_n(x)\| \le \epsilon$ for all x and $n \ge N$. ■

Corollary 6.3 (6.1.3, Weierstrass M-test)

If $|f_n(x)| \le M_n$ on X and $\sum M_n < \infty$, then $\sum f_n$ converges absolutely and uniformly on X.

Proof. For $n > m$, $|\sum_{k=m+1}^{n} f_k(x)| \le \sum_{k=m+1}^{n} M_k$, which is small uniformly in x; apply Theorem 6.2 to the partial sums. ■

Example 6.2

$\sum 2^{-n}\cos(3^n x)$ converges uniformly, so it is continuous. Weierstrass's function $\sum a^n \cos(b^n \pi x)$ ($0 < a < 1$, $ab > 1 + 3\pi/2$, b odd) is continuous but nowhere differentiable. Also, there is a continuous curve filling a square (Peano), obtained as a uniform limit.

6.2 Differentiation and integration of limits

Theorem 6.4 (6.2.1)

Let $f_n : [a,b] \to \mathbb{R}$ satisfy:
(a) each f_n is differentiable;
(b) $(f_n(x_0))$ converges for some $x_0 \in [a,b]$;

(c) (f_n') converges uniformly. Then (f_n) converges uniformly to a differentiable f, and $f'(x) = \lim f_n'(x)$.

Proof. By the mean value theorem applied to $f_m - f_n$,

$$|(f_m - f_n)(x) - (f_m - f_n)(t)| \le |x - t| \, \|f_m' - f_n'\|_{\sup}. \qquad (*)$$

With $t = x_0$ this shows that (f_n) is uniformly Cauchy, hence converges uniformly to some f. Fix x and set $\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}$, $\phi(t) = \frac{f(t) - f(x)}{t - x}$ for $t \neq x$. By $(*)$, (ϕ_n) is uniformly Cauchy on $[a, b] - \{x\}$, so $\phi_n \to \phi$ uniformly there. Let $g = \lim f_n'$. Given ϵ, choose N with $|\phi_N - \phi| < \epsilon$ on $[a, b] - \{x\}$ and $|f_N'(x) - g(x)| < \epsilon$, and δ with $|\phi_N(t) - f_N'(x)| < \epsilon$ for $0 < |t - x| < \delta$. Then $|\phi(t) - g(x)| < 3\epsilon$ for such t, so $f'(x) = g(x)$. ■

Uniform convergence of (f_n) alone is not enough: $\sin(nx)/\sqrt{n} \to 0$ uniformly, but the derivatives $\sqrt{n}\cos(nx)$ diverge.

Proposition 6.5 (6.2.2)

If Riemann integrable $f_n : [a, b] \to \mathbb{R}$ converge uniformly to f, then f is integrable and $\int_a^b f = \lim \int_a^b f_n$.

Proof. Let $\epsilon_n = \|f_n - f\|_{\sup}$. From $f_n - \epsilon_n \le f \le f_n + \epsilon_n$, the upper and lower integrals of f lie within $\epsilon_n(b - a)$ of $\int f_n$; so they differ by at most $2\epsilon_n(b - a) \to 0$, and $|\int f - \int f_n| \le \epsilon_n(b - a)$. ■

Example 6.3

Integrating $\frac{1}{1+t^2} = \sum_{n \ge 0}(-1)^n t^{2n}$ (uniform on $[-r, r]$, $r < 1$) gives $\arctan x = \sum (-1)^n x^{2n+1}/(2n + 1)$ for $|x| < 1$; similarly $\log(1 + x) = \sum (-1)^{n+1} x^n/n$. If $a_n > 0$ decrease and $\sum a_n < \infty$, then $na_n \to 0$: $\frac{n}{2} a_n \le a_{\lceil n/2 \rceil} + \cdots + a_n \to 0$.

6.3 Double sequences and double series

Proposition 6.6 (6.3.1)

Let (a_{mn}) be a double sequence such that (a) $a_{mn} \to b_m$ as $n \to \infty$ for each m;
(b) $a_{mn} \to c_n$ as $m \to \infty$ for each n;
(c) the convergence in (b) is uniform in n: for every ϵ there is N with $|a_{mn} - c_n| < \epsilon$ for $m \ge N$ and all n. Then (b_m) and (c_n) converge to the same limit: $\lim_m \lim_n a_{mn} = \lim_n \lim_m a_{mn}$.

Proof. Regard $f_m(n) = a_{mn}$ as functions on $X = \{1, 1/2, 1/3, \dots\} \cup \{0\}$ (with $n \leftrightarrow 1/n$), extended to 0 by $f_m(0) = b_m$; each f_m is continuous by (a), and by (c) and Theorem 6.2 the f_m converge uniformly on X (at 0 too, since $|b_m - b_k| \le \sup_n |a_{mn} - a_{kn}|$). The limit is continuous (Theorem 6.1), with value c_n at $1/n$ and $\lim b_m$ at 0; continuity at 0 gives $\lim_n c_n = \lim_m b_m$. ■

Proposition 6.7 (6.3.2)

If continuous $f_n : X \to Y$ converge uniformly to f and $x_n \to x$ in X, then $f_n(x_n) \to f(x)$.

Proof. $\|f_n(x_n) - f(x)\| \le \|f_n - f\|_{\sup} + \|f(x_n) - f(x)\|$, and f is continuous (Theorem 6.1). ■

With pointwise convergence this fails: $f_n(x) = x^n$, $x_n = 1 - 1/n$, gives $f_n(x_n) \to 1/e$ while $f(1) = 1$.

Proposition 6.8 (6.3.3)

If $\sum_n |a_{mn}| = b_m < \infty$ for each m and $\sum_m b_m < \infty$, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$, all series converging absolutely.

Proof. Each $\sum_m |a_{mn}| \le \sum_m b_m$, so the inner sums converge absolutely. Let $S_{MN} = \sum_{m\le M} \sum_{n\le N} a_{mn}$. As $N \to \infty$, $S_{MN} \to \sum_{m\le M} \sum_n a_{mn}$; as $M \to \infty$, $S_{MN} \to \sum_{n\le N} \sum_m a_{mn}$, uniformly in N, since $|S_{MN} - \sum_{n\le N} \sum_m a_{mn}| \le \sum_{m>M} b_m$. Proposition 6.6 gives the equality of the iterated limits. ■

Proposition 6.9 (6.3.4)

If $a_{mn} \ge 0$ for all m, n, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$ in $[0, \infty]$.

Proof. For all M, N, $\sum_{m\le M} \sum_{n\le N} a_{mn} \le \sum_{n\le N} \sum_m a_{mn} \le \sum_n \sum_m a_{mn}$. Letting $N \to \infty$ and then $M \to \infty$ gives $\le$; the reverse is symmetric. ■

Both sides also equal $\sup_F \sum_{(m,n)\in F} a_{mn}$ over finite sets F, and any enumeration of the double series of nonnegative terms has this sum (Proposition 2.16 of Volume I). For $(a_k)_{k\in\mathbb{Z}}$, $\sum_{k=-\infty}^{\infty} a_k$ means $\lim_{m,n\to\infty} \sum_{k=-m}^{n} a_k$ when this double limit exists.

Corollary 6.10 (6.3.5)

If $a_k \ge 0$ for all $k \in \mathbb{Z}$, then $\lim_n \sum_{-n}^{n} a_k = \lim_n \lim_m \sum_{-m}^{n} a_k = \lim_m \lim_n \sum_{-m}^{n} a_k$ in $[0, \infty]$.

Proof. All three equal $\sup_{m,n} \sum_{-m}^{n} a_k$, since the partial sums increase in both m and n. ■

Corollary 6.11 (6.3.6)

If $\sum_{k\in\mathbb{Z}} |a_k| < \infty$, the three limits in Corollary 6.10 exist and coincide.

Proof. Apply Corollary 6.10 to $a_k^{\pm} = \max(\pm a_k, 0)$, whose sums are finite, and subtract. ■

6.4 Power series

Theorem 6.12 (6.4.1, Cauchy–Hadamard)

Let $\lambda = \limsup |a_n|^{1/n}$ and $R = 1/\lambda$ ($R = \infty$ if $\lambda = 0$, $R = 0$ if $\lambda = \infty$). Then $\sum a_n x^n$ converges absolutely for $|x| < R$ and diverges for $|x| > R$; for every $0 < r < R$ it converges uniformly on $[-r, r]$. Its sum f is differentiable on $(-R, R)$ with $f'(x) = \sum n a_n x^{n-1}$, and hence C^∞, with $a_n = f^{(n)}(0)/n!$.

Proof. $\limsup |a_n x^n|^{1/n} = \lambda |x|$; apply the root test (Proposition 2.19 of Volume I). On $[-r, r]$ use the M-test with $M_n = |a_n| r^n$, summable as $r < R$. Since $\limsup |n a_n|^{1/n} = \lambda$ ($n^{1/n} \to 1$), the derived series has the same radius and converges uniformly on $[-r, r]$; Theorem 6.4 gives $f' = \sum n a_n x^{n-1}$. Iterate. ■

Theorem 6.13 (6.4.2, re-expansion)

If $f(x) = \sum a_n x^n$ on $(-R, R)$ and $|c| < R$, then $f(x) = \sum_n \frac{f^{(n)}(c)}{n!} (x-c)^n$ for $|x - c| < R - |c|$.

Proof. $f(x)=\sum_n a_n(c+(x-c))^n=\sum_n\sum_{k\le n}a_n\binom{n}{k}c^{n-k}(x-c)^k$. Since $\sum_n\sum_k|a_n|\binom{n}{k}|c|^{n-k}|x-c|^k=\sum_n|a_n|(|c|+|x-c|)^n<\infty$, Proposition 6.8 allows the order of summation to be exchanged: $f(x)=\sum_k(\sum_n\binom{n}{k}a_nc^{n-k})(x-c)^k$, and the coefficient of $(x-c)^k$ is $f^{(k)}(c)/k!$ by termwise differentiation (Theorem 6.12). ∎

So a power series is **analytic** (locally a power series) throughout its interval of convergence. The **Cauchy product** of $\sum a_n$ and $\sum b_n$ is $\sum c_n$, $c_n=\sum_{k=0}^n a_kb_{n-k}$.

Proposition 6.14 (6.4.3, Mertens)

If $\sum a_n=A$ converges absolutely and $\sum b_n=B$, then $\sum c_n=AB$.

Proof. With partial sums A_n,B_n,C_n and $\beta_n=B_n-B$, $C_n=\sum_{k\le n}a_kB_{n-k}=A_nB+\gamma_n$, $\gamma_n=\sum_{k\le n}a_k\beta_{n-k}$. Let $\alpha=\sum|a_k|$ and $\epsilon>0$; choose N with $|\beta_m|<\epsilon$ for $m\ge N$. Then $|\gamma_n|\le\epsilon\alpha+\sum_{n-N<k\le n}|a_k||\beta_{n-k}|$, and the last sum tends to 0 as $n\to\infty$ because $a_k\to0$ and the β's are bounded. So $\gamma_n\to0$ and $C_n\to AB$. ∎

Theorem 6.15 (6.4.4, Abel)

Let $f(x)=\sum a_nx^n$ on $(-R,R)$. If $\sum a_nR^n$ converges, then $\lim_{x\to R^-}f(x)=\sum a_nR^n$.

Proof. Replacing a_n by a_nR^n we may take $R=1$; subtracting a constant, assume $s=\sum a_n=0$. With partial sums $s_n\to0$, summation by parts gives $\sum_{n\le N}a_nx^n=(1-x)\sum_{n<N}s_nx^n+s_Nx^N$, so $f(x)=(1-x)\sum s_nx^n$ for $|x|<1$. Given ϵ, choose M with $|s_n|<\epsilon$ for $n\ge M$; then $|f(x)|\le(1-x)\sum_{n<M}|s_n|+\epsilon$ for $0<x<1$, which is $<2\epsilon$ for x near 1. ∎

Corollary 6.16 (6.4.5)

If $\sum a_n=A$, $\sum b_n=B$ and their Cauchy product $\sum c_n=C$ all converge, then $C=AB$.

Proof. For $|x|<1$ the series $\sum a_nx^n$, $\sum b_nx^n$ converge absolutely, so $\sum c_nx^n=(\sum a_nx^n)(\sum b_nx^n)$ by Proposition 6.14. Let $x\to1^-$ and use Theorem 6.15 three times. ∎

Example 6.4

Abel's theorem gives $\log 2=1-\frac12+\frac13-\cdots$ and $\frac{\pi}{4}=1-\frac13+\frac15-\cdots$ from Example 6.3. Defining $e^x=\sum x^n/n!$, Proposition 6.14 gives $e^{x+y}=e^xe^y$; with $\sin,\cos$ defined by their series, $(\sin)'=\cos$, $(\cos)'=-\sin$, and $\sin^2+\cos^2=1$ follow by termwise differentiation.

6.5 Trigonometric series

A complex vector space V with a map $\langle\cdot,\cdot\rangle$ that is linear in the first variable, satisfies $\langle y,x\rangle=\overline{\langle x,y\rangle}$, and $\langle x,x\rangle>0$ for $x\neq0$, is an **inner product space**; $\|x\|=\langle x,x\rangle^{1/2}$.

Theorem 6.17 (6.5.1, Cauchy–Schwarz)

$|\langle x,y\rangle|\le\|x\|\|y\|$ in a complex inner product space; hence $\|x+y\|\le\|x\|+\|y\|$.

Proof. If $y\neq0$, put $\lambda=\langle x,y\rangle/\|y\|^2$; then $0\le\|x-\lambda y\|^2=\|x\|^2-|\langle x,y\rangle|^2/\|y\|^2$. Then $\|x+y\|^2=\|x\|^2+2\,\mathrm{Re}\langle x,y\rangle+\|y\|^2\le(\|x\|+\|y\|)^2$. ∎

Summation by parts: with $F_n = \sum_{k\le n} f_k$ ($F_0 = 0$),

$$\sum_{k=m}^{n} f_k g_k = F_n g_n - F_{m-1} g_m + \sum_{k=m}^{n-1} F_k (g_k - g_{k+1}). \tag{$\dagger$}$$

Theorem 6.18 (6.5.2, Dirichlet–Dedekind test)

Let $f_n, g_n : X \to \mathbb{C}$ satisfy:
(a) the partial sums F_n of $\sum f_n$ are uniformly bounded, $|F_n(x)| \le K$;
(b) $g_n \to 0$ uniformly;
(c) $\sum_n |g_n(x) - g_{n+1}(x)|$ converges uniformly on X. Then $\sum f_n g_n$ converges uniformly on X.

Proof. By ($\dagger$), $|\sum_{k=m}^{n} f_k g_k| \le K|g_n| + K|g_m| + K\sum_{k=m}^{n-1}|g_k - g_{k+1}|$, which is uniformly small for large m; apply Theorem 6.2. ∎

Corollary 6.19 (6.5.3)

If the partial sums of $\sum a_n$ are bounded and b_n decreases to 0, then $\sum a_n b_n$ converges.

Proof. Here $\sum |b_n - b_{n+1}| = b_1$; apply Theorem 6.18 with constant functions. ∎

This contains the alternating series test (Theorem 2.17 of Volume I). For $x \notin 2\pi\mathbb{Z}$,

$$\sum_{k=1}^{n} e^{ikx} = e^{ix}\frac{e^{inx} - 1}{e^{ix} - 1}, \qquad \Big|\sum_{k=1}^{n} e^{ikx}\Big| \le \frac{1}{|\sin(x/2)|}, \tag{$\ddagger$}$$

so the partial sums of $\sum \sin kx$ and $\sum \cos kx$ are bounded by $1/\sin(\delta/2)$ on $[\delta, 2\pi - \delta]$.

Theorem 6.20 (6.5.4)

If $a_n \to 0$ and $\sum |a_n - a_{n-1}| < \infty$, then for every $\delta \in (0, \pi)$ the series $\sum a_n \sin nx$ and $\sum a_n \cos nx$ converge uniformly on $[\delta, 2\pi - \delta]$.

Proof. Theorem 6.18 with $f_n(x) = e^{inx}$ (bounded partial sums by ($\ddagger$)) and constant $g_n = a_n$; take real and imaginary parts. ∎

Theorem 6.21 (6.5.5)

Let a_n decrease to 0. Then $\sum a_n \sin nx$ converges uniformly on $\mathbb{R}$ iff $na_n \to 0$.

Proof. ($\Rightarrow$) Given ϵ, uniform convergence gives $|\sum_{n/2\le k\le n} a_k \sin kx| < \epsilon$ for all x and all large n. Take $x = \pi/(2n)$: for $n/2 \le k \le n$, $kx \in [\pi/4, \pi/2]$, so $\sin kx \ge 1/\sqrt{2}$, and $a_k \ge a_n$. There are at least $n/2$ such k, so $\frac{n}{2} a_n/\sqrt{2} < \epsilon$. Hence $na_n \to 0$.

($\Leftarrow$) Since a_k decreases, ($\dagger$) with $f_k = \sin kx$ and ($\ddagger$) give $|\sum_{k=m}^{n} a_k \sin kx| \le 2a_m/\sin(x/2) \le 2\pi a_m/x$ for $0 < x \le \pi$ (using $\sin(x/2) \ge x/\pi$). Let $\epsilon_m = \sup_{k\ge m} ka_k \to 0$. By oddness and periodicity it suffices to bound $R_m(x) = \sum_{k\ge m} a_k \sin kx$ for $x \in (0, \pi]$; let N be the largest integer $\le \pi/x$. If $m > N$, then $m \ge N + 1 > \pi/x$, so $|R_m(x)| \le 2\pi a_m/x < 2ma_m \le 2\epsilon_m$. If $m \le N$, split $R_m = \sum_{k=m}^{N} + \sum_{k>N}$. In the first sum $|a_k \sin kx| \le ka_k x \le \epsilon_m x$, and there are at most $N \le \pi/x$ terms, so it is at most $\pi\epsilon_m$; the second is at most $2\pi a_{N+1}/x < 2(N+1)a_{N+1} \le 2\epsilon_m$. So $\sup_x |R_m(x)| \le (\pi + 2)\epsilon_m \to 0$. ∎

Theorem 6.22 (6.5.6, Abel's test)

Let $f_n : X \to \mathbb{C}$ and $g_n : X \to \mathbb{R}$ satisfy:
(a) $\sum f_n$ converges uniformly;
(b) $g_{n+1}(x) \leq g_n(x)$ for all x, n;
(c) (g_n) is uniformly bounded, $|g_n| \leq M$. Then $\sum f_n g_n$ converges uniformly.

Proof. Apply (†) to the tails: with $R_k = \sum_{j=m}^{k} f_j$ ($R_{m-1} = 0$), $\sum_{k=m}^{n} f_k g_k = R_n g_n + \sum_{k=m}^{n-1} R_k(g_k - g_{k+1})$. Given ϵ, for m large $|R_k(x)| < \epsilon$ for all $k \geq m$, x, by (a). Since $g_k - g_{k+1} \geq 0$, the sum is at most $\epsilon M + \epsilon(g_m - g_n) \leq 3M\epsilon$ in absolute value. ■